

Nathaniel Weidner

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EDUCATION

Ohio University | Athens, OH
Bachelor of Science in Mechanical Engineering
Certificate in Project Management
Ohio Honors Program

Expected Graduation: May 2027
GPA: 3.977

EXPERIENCE

dormakaba Americas | Indianapolis, IN
Mechanical Engineering Intern

May 2026 – August 2026

- Designed a machine-guarding system for a torsion testing machine, verified to Performance Level e (PL e), with dual-channel emergency stops and safety-interlocked access
- Applied ISO, ANSI, and OSHA standards throughout the design process, translating regulatory and safety requirements into system specifications, risk mitigation measures, and validation criteria
- Created detailed Creo models, assemblies, engineering drawings, and technical documentation, including kinematic analyses, BOMs, DFMEAs, SRDs, and design review records

Ohio University | Athens, OH

October 2024 – May 2026

Mechanical Design Lead, Animatronic Project

- Led the mechanical design and development of a fully functional Gizmo animatronic, creating original components and assemblies from scratch in SolidWorks as part of an engineering team
- Designed servo-driven mechanisms for coordinated head, eye, ear, arm, and leg motion while integrating 3D-printed and commercial components to achieve the required movement and functionality
- Led mechanical integration, troubleshooting, and iterative design improvements throughout prototyping, assembly, and testing to resolve motion and fit issues and complete the functional system

Ohio Department of Natural Resources | Cambridge, OH

May 2024 – August 2024

Abandoned Mine Land Engineering Intern

- Developed AutoCAD engineering plan sets for abandoned mine land reclamation projects, incorporating aerial mapping, contour data, site layouts, and detailed construction drawings
- Prepared technical specifications, analyzed site data, and supported design reviews and field inspections for projects addressing hazardous mine openings and public safety concerns

PROJECTS

Modular Deployable Rocket Payload | Ohio University Rocket Club

August 2026 – Present

Senior Design Project

- Designing a modular deployable rocket payload for interchangeable equipment such as cameras, environmental sensors, and IMUs while meeting defined mass, packaging, and integration constraints
- Developing and evaluating deployment concepts that keep the payload contained within the rocket during ascent while allowing safe deployment, recovery, and repeatable ground testing
- Translating customer, safety, and flight requirements, including launch loads up to 30 g, into mechanical design criteria used to compare concepts, guide development, and support design decisions

LEADERSHIP AND INVOLVEMENT EXPERIENCE

Tau Beta Pi | Athens, OH

November 2024 – April 2026

Vice President

- Helped coordinate biweekly chapter meetings, social events, and new-member induction while supporting chapter activities and member engagement

Ohio University | Athens, OH

October 2023 – May 2026

Mechanical Engineering Department Student Liaison

- Collaborated with Mechanical Engineering faculty and fellow student liaisons to communicate student feedback, support department events, and identify opportunities to improve the student experience

TECHNICAL SKILLS

CAD & Design: SolidWorks, Creo Parametric, AutoCAD, Engineering Drawings

Programming & Analysis: MATLAB, Python, Excel

Design & Manufacturing: 3D Printing, GD&T Fundamentals, Prototyping, BOM Development, DFMEA